INCH-POUND

MIL-S-22820/34C
28 November 1990
SUPERSEDING
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MILITARY SPECIFICATION SHEET
SERVOMOTOR-TACHOMETER GENERATOR, TYPE 15SM-1H4A (HEAT COMPENSATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Servomotor-Tachometer Generators described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-S-22820.

TABLE I. Requirements of servomotor section.

Requirement	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	hz	
Test	400	hz	±1%
Range	400	hz	±5%
Voltage:			
Reference phase, rated	115	volts	
Reference phase, test	115	volts	±1%
Reference phase, range	103 to 126	volts	
Control phase, rated (series)	115	volts	
Control phase, test (series)	115	volts	±1%
Control phase, range (series)	0 to 126	volts	
Control phase, rated (parallel)	57.5	volts	±1%
Control phase, range (parallel)	0 to 66	volts	
Current:			
Reference phase	0.140	amperes	maximum
Control phase (series)	0.140	amperes	maximum
Control phase (parallel)	0.280	amperes	maximum
Power:			
Reference phase	8.1	watts	maximum
Control phase (series)	8.1	watts	maximum
Control phase (parallel)	8.1	watts	maximum
Impedance:			
Reference phase	560+j980	ohms	±30%
Control phase (series)	560+j980	ohms	±30%
Control phase (parallel)	140+j240	ohms	±30%
Power factor:			
Reference phase	0.48		±0.10
Control phase	0.48		±0.10
Electrical breakaway torque	2.0	volts	maximum
Coupling in control winding	0.8	ratio	minimum

AMSC - N/A

FSC 6105

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TABLE I. Requirements of servomotor section - Continued.

Requirements	Value	Unit	Tolerance
Mechanical:			
No-load speed	4,500	rpm	minimum
Stall torque	1.45	ounce-inches	minimum
Stall torque linearity	10.0	percent	maximum
Rotor moment of enertia	10.0	$gm-cm^2$	maximum
Radial play	0.0008	inches	maximum
End play	0.0010-0.0045	inches	min -max.
Temperature rise	85.0	degrees C	maximum
Maximum power output	1.2	watts	minimum
Weight	16.0	ounces	maximum

TABLE II. Requirements for tachometer-generator section.

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Test	400	Hz	±1%
Range	400	Hz	±5%
Voltage:			
Energizing winding, test	115	volts	±1%
Energizing winding, range	103 to 126	volts	min.-max.
Current:			
Energizing winding	0.09	amperes	maximum
Power:			
Energizing winding	7.0	watts	maximum
Power factor:			
Energizing winding	0.60	-----	±0.1
Impedance:			
Energizing winding	945+j1300	ohms	±30%
1/In-phase speed-sensitive transformation ratio (at 3000 rpm) 2/	0.07174	ratio	±0.5%
1/Phase shift (at 3000 rpm) 2/	0.5	degrees	maximum
1/In-phase linearity 2/	0.07	percent	maximum
1/In-phase axis error 2/	0.0035	volts	maximum
1/Quadrature-phase axis error 2/	0.007	volts	maximum
1/In-phase position error 2/	0.010	volts	maximum
1/Quadrature-phase position error 2/	0.020	volts	maximum
1/Zero-speed output voltage 2/			
Total	0.025	volts	maximum
Fundamental	0.020	volts	maximum

1/At 70°C ± 0.1%

2/Measured open circuit for values specified in Table II.

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TABLE II. Requirements of tachometer generator section - Continued.

Requirements	Value	Unit	Tolerance
In-phase speed-sensitive output at 3000 rpm (temperature variation 60°C to 80°C from value at 70°C)	0.05	percent	maximum
Zero degrees C to 80°C	0.4	percent	maximum
In-phase axis-error (maximum variation 60°C to 80°C from value at 70°C)	0.005	volts	maximum
Quadrature-phase axis-error (maximum variation 60°C to 80°C from value at 70°C)	0.005	volts	maximum

TABLE III. Part identifying number variant characteristics.

Part Identifying Number 1/	Shaft Style	Termination	G
M22820/34-01A	Splined	Terminal board	4.00

1/Part identifying number M22820/1-01A shall be the NATO standard.

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE IV. Part identifying number cross references.

SUPERSEDED PART NUMBER	NEW PART IDENTIFYING NUMBER
BUORD Drawing 2073896	
2073896	M22820/34-01A

1/All the line items shown in Table IV refer to equivalent and interchangeable servomotor-tachometer generators of the same type designation modification. Part number changes do not affect form, fit or function of the servomotor-tachometer generators listed therein.

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NOTES:

1. Type 15SM-1H4A (heat compensated) servomotor-tachometer generators shall be capable of operation in ambient temperatures ranging from 0° to 80°C.

2. The speed range for the 15SM-1H4A is 0 to 4,500 rpm. The calibration speed for linearity requirements shall be 3,000 rpm.

3. All electrical tests are to be conducted with the generator output, terminals 8 and 9, connected to a 100K ohm ±.05% resistive load.

Custodian:
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